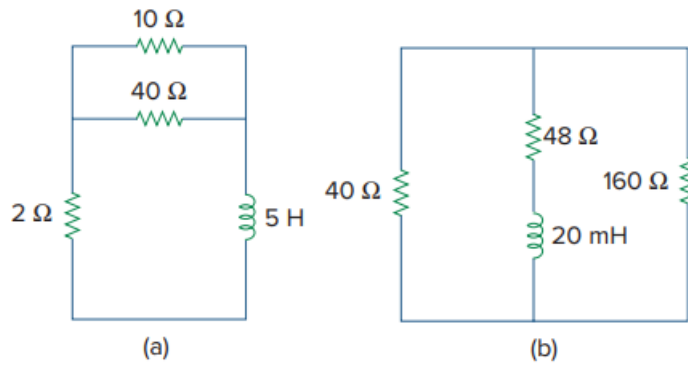


Problem

Find the time constant for each of the circuits in Fig. 7.95.

Fig. 7.95:



Step-by-step solution

Step 1 of 5

(a)

Consider the following circuit diagram:

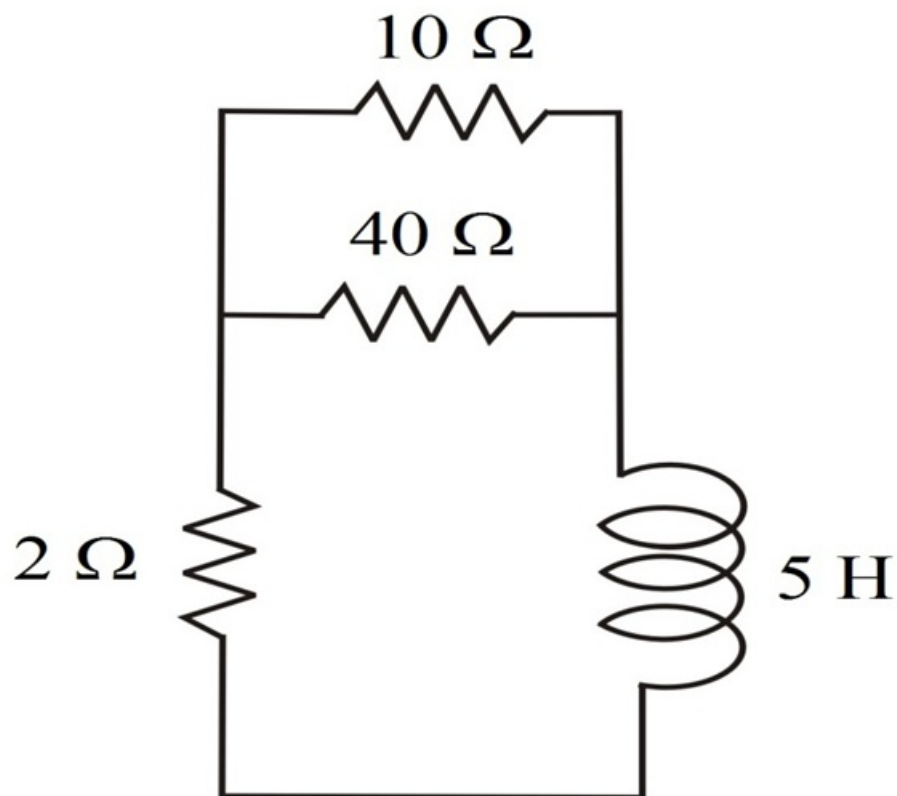


Figure 1

Step 2 of 5

Write an expression for time constant of RL circuit.

$$\tau = \frac{L}{R_{eq}}$$

R_{eq} is equivalent resistance

Looking from the terminals of the inductor, resistors $40\ \Omega$ and $10\ \Omega$ shown in the Figure 1 are connected in parallel, and this parallel resistance is in series with the $2\ \Omega$ resistor. Calculate the equivalent resistance.

$$\begin{aligned} R_{eq} &= [(40) \parallel (10)] + 2 \\ &= \frac{(40)(10)}{40+10} + 2 \\ &= \frac{400}{50} + 2 \\ &= 8 + 2 \\ &= 10\ \Omega \end{aligned}$$

Step 3 of 5

Calculate the time constant.

$$\begin{aligned} \tau &= \frac{L}{R_{eq}} \\ &= \frac{5}{10} \\ &= \frac{1}{2}\text{ s} \end{aligned}$$

Hence, the time constant, τ of RL circuit is $\boxed{\frac{1}{2}\text{ s}}$.

Step 4 of 5

(b)

Consider the following circuit diagram:

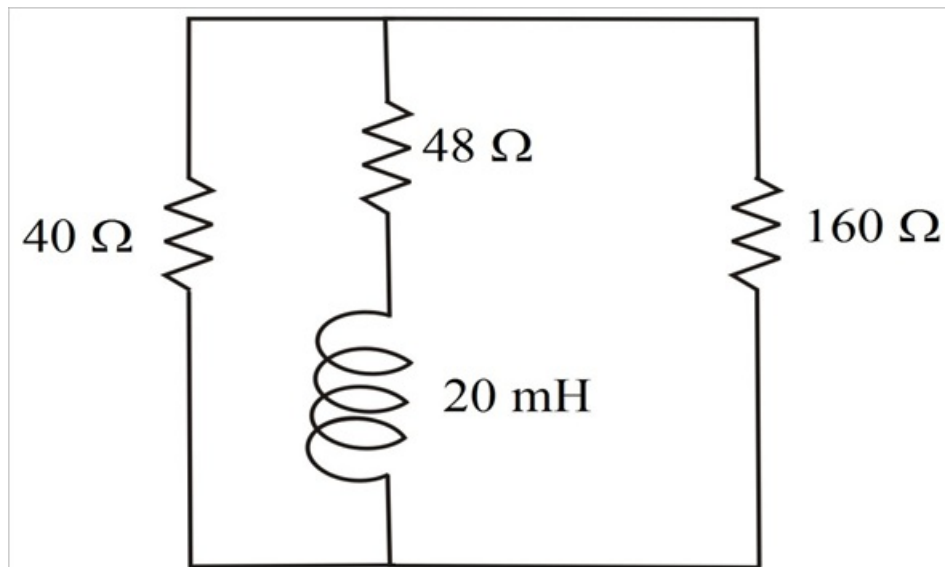


Figure 2

Step 5 of 5

Looking from the terminals of the inductor, resistors $40\ \Omega$ and $160\ \Omega$ shown in the Figure 2 are connected in parallel, and this parallel resistance is in series with the $48\ \Omega$ resistor.

Calculate the equivalent resistance.

Calculate the equivalent resistance.

$$\begin{aligned} R_{eq} &= [(40) \parallel (160)] + 48 \\ &= \frac{(40)(160)}{40 + 160} + 48 \\ &= \frac{6400}{200} + 48 \\ &= 32 + 48 \\ &= 80\ \Omega \end{aligned}$$

Calculate the time constant.

$$\begin{aligned} \tau &= \frac{L}{R_{eq}} \\ &= \frac{20 \times 10^{-3}}{80} \\ &= 250\ \mu s \end{aligned}$$

Hence, the time constant, τ of RL circuit is $250\ \mu s$.